

RAJKUMAR SHANMUGAM

Software Engineer — Security-Focused Systems & Applications

Chennai, India | +91 8925615811 | rajkumarshanmugam151@gmail.com

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Medium](#)

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building network management systems, CLI/SNMP-based automation, and full-stack applications using Java (Spring Boot), React, and Python, with a strong secondary specialization in application and network security. Delivered a multi-vendor NMS platform, contributed to SONiC NOS BGP CLI development, and integrated Wazuh SIEM, IAM (OKTA/Auth0), and OWASP-aligned security practices into engineering workflows. Recognized with 20+ Hall of Fame acknowledgements from NASA, BBC, WHO, and UNESCO for responsible vulnerability disclosure.

TECHNICAL SKILLS

Languages & Frameworks: Java (Spring Boot), Python, C, React, JavaScript

Networking & NMS: SONiC NOS, BGP (vtysh / IS-CLI), SNMP, CLI-based device management, TCP/IP, VLAN, ICMP, ARP, DNS, Cisco Packet Tracer

AI & LLM Security: Claude, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), OWASP LLM Top 10, Agentic AI Security, Prompt Injection Defense

Security & IAM: OWASP Top 10, VAPT, OKTA, Auth0, OAuth 2.0, MFA, SSO, Federation, PAM, Zero Trust

SIEM & Monitoring: Wazuh (custom decoders & detection rules), real-time alerting, anomaly detection

Tools & Platforms: Burp Suite, Git/GitHub, Docker, ArangoDB, Postman

Operating Systems: Ubuntu/Linux (Kali), Windows

CORE COMPETENCIES

Cross-Functional Collaboration • Stakeholder Communication • Technical Writing & Documentation • Mentoring • Ownership & Autonomy • Team Leadership

PROFESSIONAL EXPERIENCE

Software Engineer — PalC Networks Pvt. Ltd.

Aug 2025 – Present

Chennai, India

- Built and maintained a multi-vendor Network Management System (NMS) supporting CLI and SNMP protocols using Java (Spring Boot) and React, enabling centralized monitoring and management across diverse network infrastructure.
- Integrated SONiC NOS CLI support and improved BGP CLI using the IS-CLI framework, streamlining BGP session management and fault detection on open networking platforms.
- Engineered real-time device health logging, anomaly detection, and threat visualization dashboards, layering security monitoring capabilities directly into the NMS platform.
- Conducted vulnerability assessments and security reviews alongside core development work as part of ongoing infrastructure hardening initiatives.

Project Engineer — PalC Networks Pvt. Ltd.

Jun 2023 – Jul 2025

Chennai, India

- Developed a CLI-only NMS module for devices lacking SNMP support, enabling structured, secure interaction with a wider range of network hardware.
- Simulated and troubleshooted routing, VLANs, and network protocols (ICMP, ARP, DNS) using Cisco Packet Tracer to validate network topologies before deployment.
- Hardened email security posture by configuring SPF, DKIM, and DMARC records, reducing phishing risk by 60% and mitigating spoofing attacks by 80% across client domains.

- Customized Wazuh SIEM with tailored decoders and detection rules for SONiC NOS environments, enabling real-time alerting on BGP route changes and unauthorized configuration modifications.
- Identified and remediated information leakage and security misconfigurations in WordPress environments using Burp Suite and manual testing, aligned to OWASP Top 10.

Software Intern (Security) — PalC Networks Pvt. Ltd.

Feb 2023 – Jun 2023

Chennai, India

- Implemented IAM solutions using OKTA, Auth0, and OAuth 2.0, enforcing zero-trust access controls across enterprise systems.
- Deployed MFA, SSO, Federation, and Privileged Access Management (PAM) to strengthen authentication and access-control policies.
- Supported credential lifecycle management and identity governance policy enforcement.

CERTIFICATIONS

- **Certified API Pentesting & Bug Bounty 101** | Capprico Security University | May 2026 | [Verify](#)
- **Certified WiFi Pentesting 101** | Capprico Security University | May 2026 | [Verify](#)
- **Network Analysis** | Security Blue Team | May 2023 | [Verify](#)
- **Cybersecurity Virtual Experience Program** | Mastercard (Forge) | Feb 2023 | [Verify](#)

EDUCATION

B.E. in Computer Science — Mepco Schlenk Engineering College, Sivakasi

Sept 2019 – May 2023 | CGPA: 7.6 / 10

SECURITY RESEARCH & NOTABLE FINDINGS

- Identified and responsibly disclosed an email security misconfiguration that allowed unauthorized, spoofed outbound email delivery.
- Discovered a payment race-condition vulnerability enabling ₹1 completion of high-value transactions and unlimited repeated actions, creating a denial-of-service risk.
- Found a stored XSS vulnerability in a community chat's comment feature that could be chained to achieve full account takeover.
- Identified Broken Object-Level Authorization (BOLA) across multiple production APIs, exposing sensitive user data at critical endpoints.
- Uncovered sensitive information leakage in a delivery-service link through VirusTotal-based reconnaissance on a private bug bounty program.

ACHIEVEMENTS & RECOGNITION

- 20+ Hall of Fame recognitions from NASA, BBC, WHO, UNESCO, and other global organizations for responsible vulnerability disclosure.
- Top 15 National Ranking — NCPIII Q3 2024, Cybersecurity Excellence category.
- “Extra Miller” Award — PalC Networks, for outstanding performance beyond defined role.
- Active security researcher and bug bounty contributor on YesWeHack.

LANGUAGES

Tamil (Native), English (Upper Intermediate)